



Francisco Javier Guinot Almenar

Software Engineer | Game Engine Developer | Low-Level Programmer

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Skills

📁 Languages

Advanced

C, C++, C#, Python, SQL, GLSL, Bash /
Shell Scripting, Batch Scripting, x86/x64 Assembly
(Reading/Debugging)

🔧 Computer architecture

Advanced

Cache-friendly programming, Memory coherency, Data
alignment (SIMD), Cache miss reduction

🔗 Graphics & Low-Level

Advanced

Vulkan, OpenGL, Multi-threading (Lock-free primitives),
Engine Development, Real-time Simulation, Custom
memory allocators, Real-time simulation

🔧 Hardware & systems

Advanced

Peripheral communication protocols (USB / Industrial
protocols), Real-time operating systems (RTOS)
concepts, Linux/Windows internals

📦 Tools & Frameworks

Advanced

Git, CMake, Visual Studio, Unity, Unreal Engine 5, Jira,
Subversion (SVN), RenderDoc, NVIDIA NSight (Profiling)

Languages

Spanish

Native

English

Full professional proficiency

Interests

Low-level hardware architecture

Deep interest in understanding CPU/GPU internals -
cache hierarchy and how silicon executes code

Retro-computing & emulation

Developing and studying emulators to understand how
legacy hardware managed limited resources through
extreme optimization

Mathematics for compute graphics

Constant learning of linear algebra - calculus and
numerical methods applied to physics and rendering

Physics simulations

Exploring GPGPU techniques to simulate complex
natural phenomena (fluids - soft bodies and
thermodynamics) in real-time

Summary

Performance-driven software engineer specializing in high-fidelity simulation and graphics engine development. Expert in modern C++, manual memory management, and hardware-level optimization. Extensive experience building graphics engines from scratch (Vulkan/OpenGL) and real-time simulators. Focused on Data-Oriented-Design (DOD) and high-efficiency architectures (ECS) to maximize CPU/GPU throughput and ensure deterministic, low-latency systems.

Technical Philosophy

I advocated for minimal-dependency development to maintain total control over the software lifecycle. I prioritize "hardware-aware" code, understanding that every software decision from smart vs. raw pointers to data layout in memory directly impacts the latency and efficiency of the final system.

Experience

Simmodar Training Solutions (LSyM - University of Valencia) Burjassot, Valencia
Software Developer / Systems Programmer Oct 2024 - Present

- Developed and optimized high-performance backends for professional-grade simulators (Heavy machinery, maritime and logistic systems).
- Engineered core simulation logic and real-time physics using C++ and PhysX5 with a strict focus on low latency response and physical accuracy.
- Hardware-in-the-loop (HIL): Developed communication layers to synchronize physical cabins and sensors with the simulation engine.
- Optimized rendering pipelines to ensure stable frame rates in high-complexity environments, managing resources and memory to prevent bottlenecks
- Utilized professional version control and project management workflows (SVN / Jira / Mantis)

[LSyM Web](#)

Projects

Custom OpenGL Game Engine (ECS & Deferred rendering) Sep 2023 - Jun 2024

- Architected a 3D engine from the ground up focusing on performance and scalability
- ECS (Entity component system): Designed a data-oriented framework to optimize cache locality (L1/L2) and handle massive entity processing efficiently.
- Deferred rendering: Implemented a multi-pass deferred shading pipeline, optimizing complex lighting calculations and scene depth management.
- Multithreading: Engineered an asynchronous resource loading system and parallelized engine subsystems to prevent main-thread bottlenecks.
- Instanced rendering: Leveraged GPU instancing to minimize draw calls for massive geometry rendering.
- [GitHub project \(Public\)](#)

Real-Time Fluid simulator (Voxel-Based Spatial partitioning) Sep 2023 - Jun 2024

- Developed a large-scale particle simulator capable of processing hundreds of thousands of particles in real-time.
- Implemented a Voxel-based neighbour search algorithm, significantly reducing computational complexity.

- Solved floating-point precision challenges and thread synchronization issues in parallel GPU computing environments.

Vulkan Graphics Engine (Cross-Platform)

(Ongoing)

- Ongoing development of a graphics engine designed for maximum hardware control (Windows / Linux / Android)
- Manual management of Vulkan memory allocator, synchronization primitives (fences / semaphores), and command queues.

Education

ESAT (Escuela Superior de Arte y Tecnología)

HND in Computing and Systems Development

Game Programming

Valencia • Sep 2021 - Jun 2024

- Intensive focus on **Advanced C++**, Data Structures, Computer Architecture and Gaming engines (Unity / Unreal Engine 5)
- Developed custom game engines from scratch using C++ and Graphics APIs (OpenGL).
- Gained a deep understanding of memory management, optimization, and software engineering patterns applied to real-time systems.

[E.S.A.T Web](#)

Personal webs

[Linkedin](#)

[Github](#)

Portfolio: gitjvguinot.github.io